6.6 Integration by Parts

Name _____ Period ____ Date _____

AP Calculus 6.6 (30-minute plan) · Student sheet

I can…

- choose u and dv for integration by parts.
- use integration by parts, more than once if needed.
- evaluate a definite integral by parts.

Vocabulary · fill in the missing word

Integration by parts — $\int u\,dv = uv - \int v\,du$, the Product Rule run _____

Part 1 · Notes and examples (fill in each note, then work the example)

Integration by parts: $\int u\,dv =$ _____ $- \int v\,du$.

Example 1

Find $\int x \cos x\,dx$.

Choosing u: pick u so that du is _____ than u (logs and inverse trig first, then powers of x).

Example 2

Find $\int \ln x\,dx$.

Repeat: if the new integral still has a product, use parts _____.

Definite integrals: $\int$ from a to b of u dv = [uv] from a to b $-$ $\int$ from a to b of v du; evaluate uv at the _____.

Example 3

Evaluate $\int_0^1 xe^x\,dx$.

Part 2 · Partner practice

Partner practice

(AP® Practice) $\int x \sin x \, dx =$

(A) $-x \cos x + \sin x + C$ (B) $x \cos x - \sin x + C$

(C) $-x \cos x - \sin x + C$ (D) $\left(\frac{x^2}{2}\right)\sin x + C$